

ID:

Name:

Brac University

Semester: Fall 2025

Course Code: CSE251

Electronic Devices and Circuits

Set

A

Assessment: *Midterm*

Duration: 1 hour 30 minutes

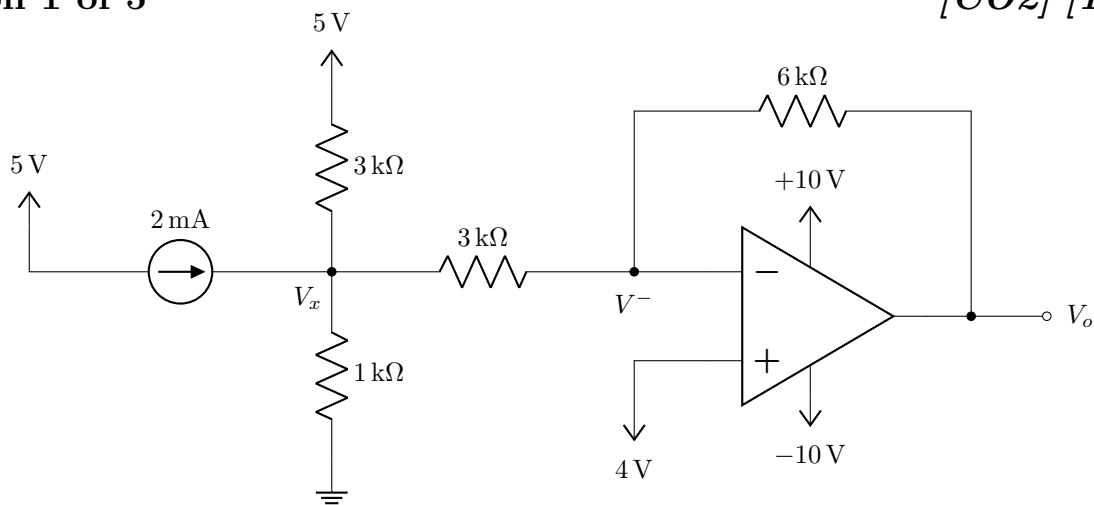
Date: November 22, 2025

Full Marks: 55

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ **All 5 question(s) are compulsory.** Marks allotted for each question are mentioned beside each question.
- ✓ Write your answers inside the indicated boxes where applicable.
- ✓ For every problem involving an operational amplifier (op-amp), consider it to be ideal.
- ✓ All calculated values must be accompanied by appropriate units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 5

[CO2] [10 marks]



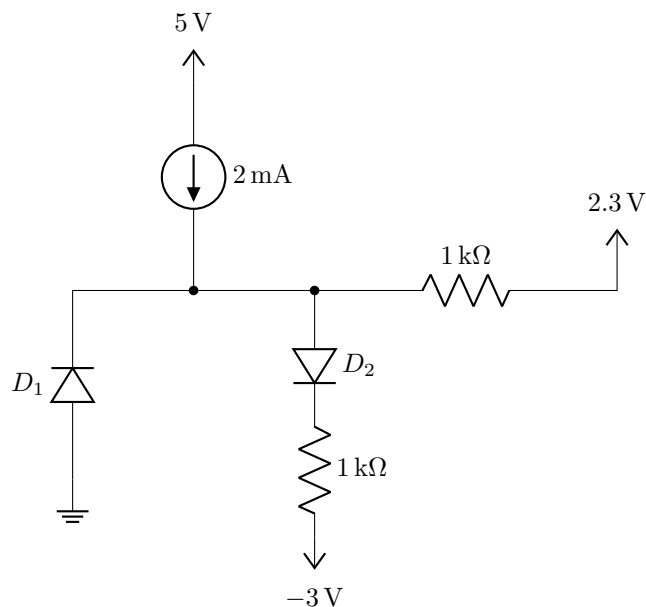
Analyze the circuit shown above and answer the following queries—

- (a) [1 mark] What is the value of V^- ?
- (b) [5 marks] Determine V_x .
- (c) [4 marks] Determine V_o .

■ Question 2 of 5

[CO2] [12 marks]

Assuming a cut-in voltage of 0.7 V for the identical diodes D_1 and D_2 in the circuit shown below, **Analyze** the circuit to determine which diode(s) are conducting. Additionally, calculate the power dissipated by any conducting diode(s).



■ Question 3 of 5

[CO3] [15 marks]

- (a) In a diode logic circuit, a logic HIGH is approximately 1.8 V, while a logic LOW is approximately 0 V. The circuit is designed to implement the Boolean function

$$f = w \cdot (x + y) \cdot z,$$

where w , x , y , and z are the Boolean inputs and f is the Boolean output.

- (i) [5 marks] **Draw** the diode logic circuit that realizes the Boolean function given above.
- (ii) [3 marks] For this logic circuit, determine the output voltage corresponding to the input values $w = 0.3$ V, $x = 0.7$ V, $y = 1.7$ V, and $z = 1.9$ V. Assume all diodes are identical with negligible resistance and have a cut-in voltage of 0.7 V.
- (b)

In the following circuit, v_{in} is a triangular voltage plotted as a function of time t in *Figure 1*, and V_{ref} is a constant DC voltage. The output voltage waveform v_o is shown on *Figure 2*.

Design the circuit to achieve the desired output behavior—

- (i) [3 marks] Determine V_{ref} and specify the biasing voltages V_{sat}^+ and V_{sat}^- .
- (ii) [4 marks] Draw the voltage transfer characteristic (VTC) of the circuit on the blank grid of *Figure 3*.

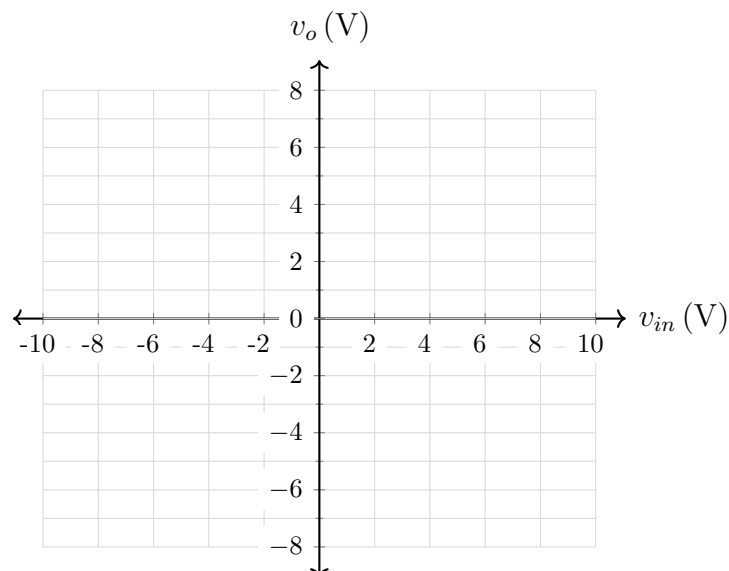
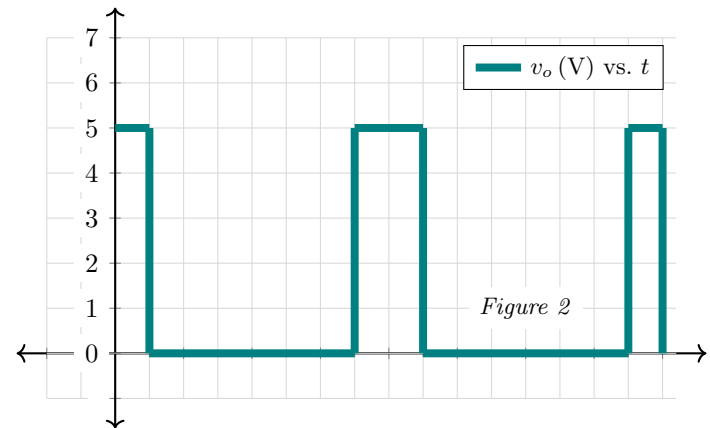
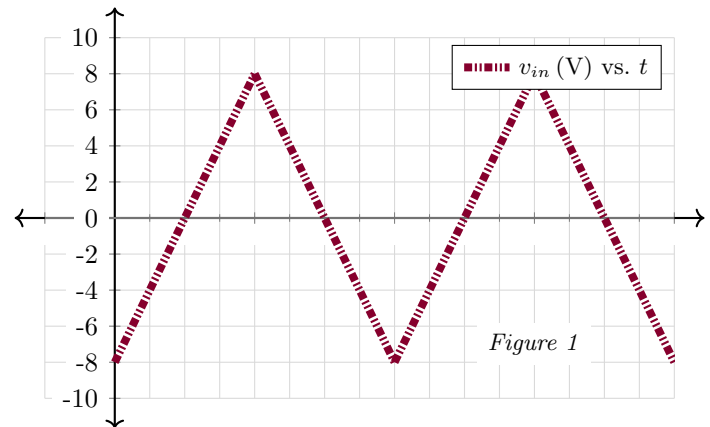
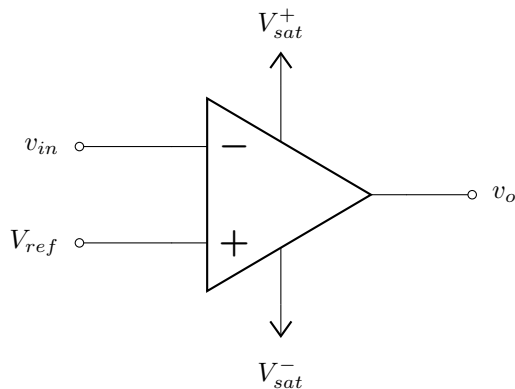
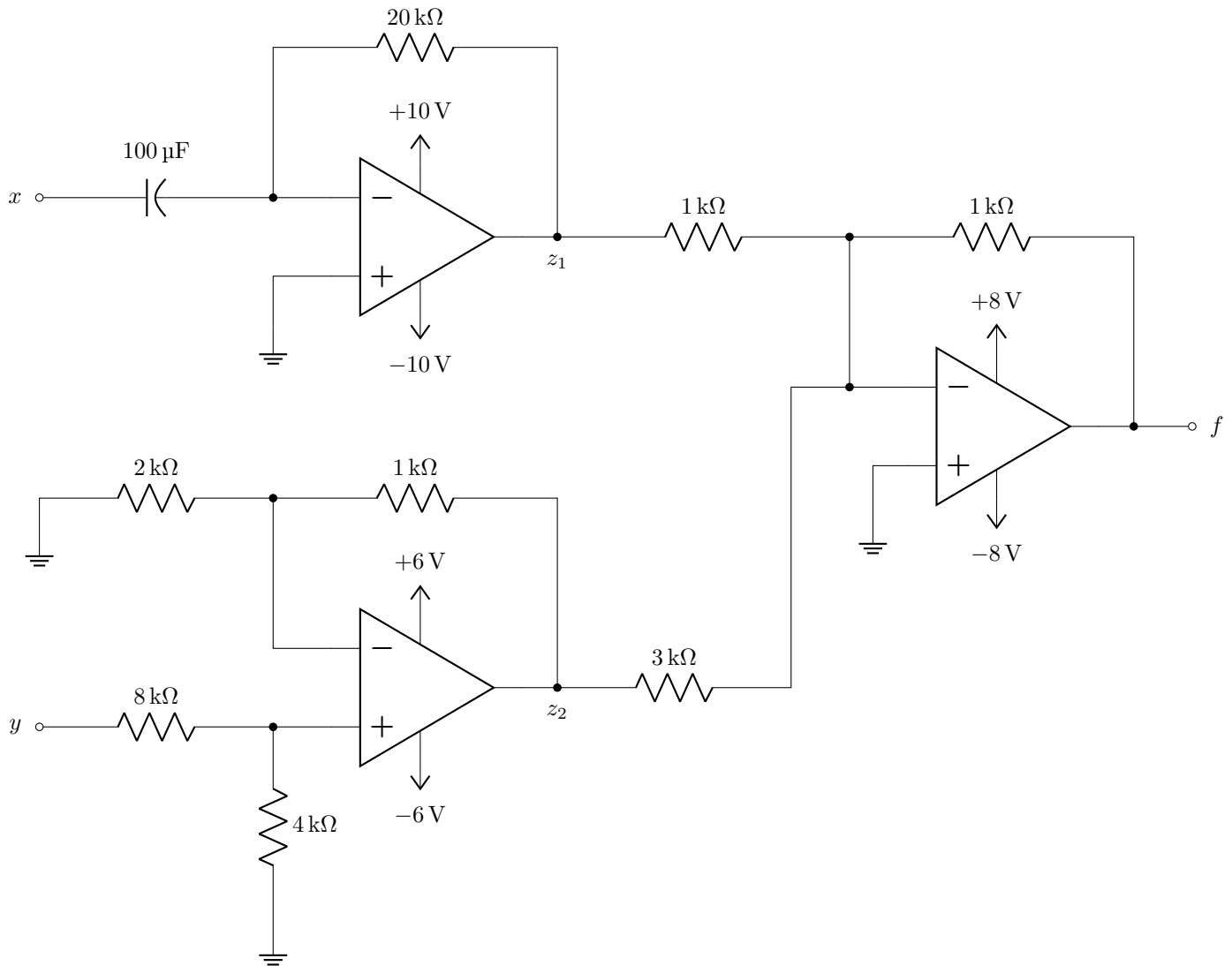


Figure 3

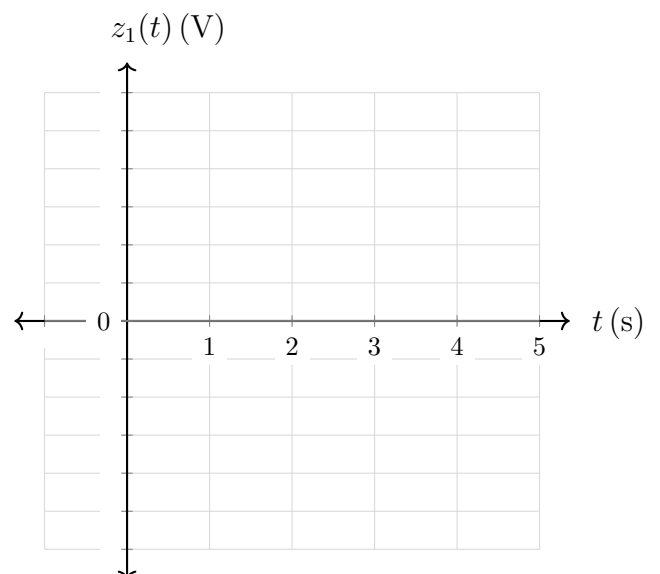
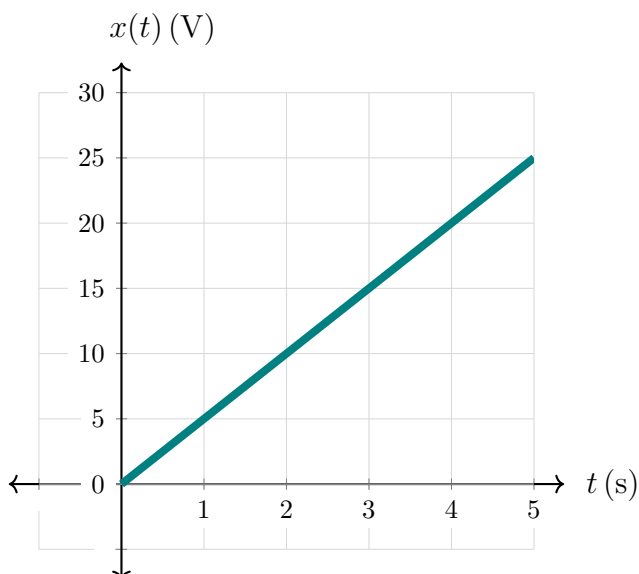
■ Question 4 of 5

[CO2] [10 marks]



Analyze the circuit shown above and answer the following queries—

- (a) [7 marks] Express f as a function of the variables x and y .
- (b) [3 marks] If x is a time-varying voltage as shown in the following plot, sketch the corresponding plot of z_1 on the blank grid below.



■ Question 5 of 5

[CO1] [8 marks]

- (a) [4 marks] **Compare** the Constant Voltage Drop (CVD) diode model with the CVD-plus-resistor model, and explain which model more accurately represents real diode behavior and why.
- (b) [2 marks] **Identify** which one of the following information of an ideal op-amp is incorrect. Provide the corrected version.
- (i) The open-loop gain is infinite.
 - (ii) The voltage difference between the inverting and non-inverting input terminals is zero regardless of the feedback configuration.
 - (iii) Output current can be non-zero.
- (c) [2 marks] **Identify** which of the following statements is true for the given connections of the identical diodes shown below. Write the correct letter in the boxes.
- A. Only one of the diodes can be ON.
 - B. Both diodes will be OFF.
 - C. Both diodes will be ON.
 - D. Either both diodes are ON or both are OFF.

